Neural Network Surrogates for TGLF: Prediction Accuracy and Uncertainty

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Quarter 1 checkpoint draft

Abstract

Repeated runs of the gyrokinetic code CGYRO and the trapped gyro-Landau fluid (TGLF) transport model can be costly. This project investigates methods for approximating the ion heat, electron heat, and electron particle fluxes predicted by TGLF. We explore a variety of TGLF simulation runs, compare candidate models, and test choices in input representation, network design, and training to produce a machine learning model comparable to TGLF. An initial study of 43 scan cases shows that greater model flexibility does not consistently improve prediction and that low aggregate error can coexist with incorrect flux signs. A comparison on a larger parameter-scan database shows that a five-network ensemble performs best among the modeling techniques, reducing test root mean squared error by 59.6%–77.2% relative to a linear baseline across the three outputs. A controlled activation comparison also reveals different tradeoffs for heat and particle flux prediction. However, uncertainty analysis shows that the accuracy of the neural network ensemble depends heavily on the TGLF data it is trained on. Test data with variables outside the range of the training data, or even data from the DIII-D tokamak that is not distributed like the training data, is subject to significant accuracy drops, usually through a systematic bias in flux predictions. Overall, this motivates further development and tuning through a larger variety of training data and continued robust uncertainty analysis.

Code: https://github.com/lindbergryan04/dsc180a-project-1

1 Introduction

1.1 Problem Statement and Research Question

Turbulence transports heat and particles across a tokamak plasma and affects its temperature and density profiles. Predicting these fluxes is part of modeling plasma confine-

ment. Detailed turbulence simulations are expensive, and even reduced models such as the trapped gyro-Landau fluid (TGLF) model can become costly when a larger calculation requires many evaluations (Meneghini et al. 2017; Cao et al. 2026). A surrogate learns to approximate a simulation from previously computed input-output pairs, making repeated predictions less expensive.

Our research question is which modeling approaches and design choices provide useful approximations of TGLF under the available data and computational constraints. We began by examining the plasma inputs, output scales, and data quality, then explored regression, tree-based, and neural network methods. We assess choices through prediction error, computational cost, and behavior across plasma conditions. Uncertainty estimates provide an additional way to examine model reliability. One case study uses plasma conditions from DIII-D, a tokamak operated by General Atomics in San Diego. A tokamak uses magnetic fields to confine hot plasma in a doughnut-shaped chamber. We use inputs based on a DIII-D experiment to compare surrogate predictions with TGLF outputs.

The prediction targets are ion heat flux Q_i, electron heat flux Q_e, and electron particle flux Γ_e. The reference values are TGLF outputs. Agreement with these values measures approximation of TGLF; it does not by itself establish agreement with experimental measurements or nonlinear gyrokinetic simulations. This checkpoint reports an exploratory comparison on a small scan dataset and a more developed neural network study on a larger database. The distinction between these experiments matters when interpreting which methods performed well.

1.2 Related Work on Neural Network Transport Surrogates

Meneghini et al. (2017) developed neural network surrogates for TGLF and for EPED1, a model of the plasma edge pedestal, and incorporated them into a coupled core and pedestal modeling workflow. Their TGLF architecture description in Section 3.1 specifies an ensemble of 32 independently initialized networks, each with one hidden layer of 60 neurons. This provides a precedent for averaging networks and a reference for investigating ensemble size. It does not establish an optimal member count for our data. Our deeper network architecture and training database differ from theirs, so our experiments evaluate the approach in a different setting.

Model evaluation also requires attention to local behavior. van de Plassche et al. (2020) developed the QuaLiKiz neural network (QLKNN) to approximate QuaLiKiz, a quasilinear gyrokinetic transport model. Their work emphasizes evaluating local transport behavior alongside aggregate prediction error, motivating our checks of flux trends and individual prediction failures. Cao et al. (2026) developed a wavenumber-informed neural network for TGLF (TGLF-WINN) using feature engineering, transformed flux targets, and supervision of flux contributions at individual wavenumbers. Their comparisons of individual design components illustrate how controlled experiments can identify useful modeling choices. We use integrated flux targets and test simpler representations and network choices; we do not implement their wavenumber supervision in the experiments reported here.

Training data selection is another direction in this literature. Zanisi et al. (2024) introduced Active Deep Ensembles for Plasma Turbulence, using neural network ensembles to guide training-data selection for QuaLiKiz surrogates. This provides context for studying data requirements and uncertainty, although our current experiments use fixed training datasets rather than active acquisition. Finally, Dudding et al. (2022) developed the third version of the TGLF saturation rule (SAT3) using nonlinear simulations with the gyrokinetic code CGYRO, with particular attention to isotope dependence. The course's starter dataset draws on that work, but the added TGLF outputs supplied for our analysis use the second version of the saturation rule (SAT2). The saturation rule and reference simulation must therefore remain explicit.

1.3 Datasets: Starter Scans, `scan_db`, and Evaluation Sets

The starter dataset contains 43 plasma cases arranged in controlled scans around a reference condition. Case names identify 26 letter groups and an ion isotope. Eight groups contain complete hydrogen, deuterium, and tritium triplets. Some inputs change together: for example, the electron and ion temperature gradients are equal in these cases. These dependencies limit the separate effects that can be inferred from correlations. We retained all 43 cases after basic validity checks, including cases flagged for further review.

The larger `scan_db` dataset contains 5,000 TGLF cases with multiple inputs varied simultaneously over prescribed ranges. The sampled continuous coordinates use uniform distributions, with collision frequency sampled logarithmically. Unlike the starter scans, electron and ion temperature gradients vary independently. Each case supplies a complete input listing in `input.tglf.gen` and species-resolved fluxes in `out.tglf.gbflux`. We retain the supplied gyro-Bohm normalization and the sign of particle flux.

Additional supplied data support focused checks. The `scan_db_outside` collection contains nine groups of 30 cases, each placing one input outside its training range. The `data_d3d` collection includes 300 TGLF evaluations with perturbed inputs around a DIII-D reference condition. These serve as evaluation cases in the uncertainty study; they are distinct from the new DIII-D training database described in the remaining work.

Each of the nine `scan_db_outside` groups sweeps exactly one input strictly beyond the range it was sampled from in `scan_db`, while holding every other input inside its nominal training range (Table 1). This design isolates extrapolation along one coordinate at a time instead of the joint extrapolation that an arbitrary new input vector could produce.

A separate 1,000-case DIII-D database was also generated, independent of `data_d3d`, following the mentor-prescribed recipe with a member-specific seed of 4. Cases are distributed evenly across the 13 supplied radii ($r/a = 0.20$ to 0.80). At each radius, five inputs are perturbed from the supplied baseline profile by independently sampled multiplicative factors: a/L_{Te} and a/L_{Ti} each by a separately drawn factor in $[0.6, 1.4]$, the shared density-gradient scale a/L_n (applied equally to both species) by a factor in $[0.6, 1.4]$, the temperature ratio T_i/T_e by a factor in $[0.75, 1.25]$, and the collision frequency ν_{ei} by a factor in $[0.5, 2.0]$ drawn uniformly on a $\log_{10}$ scale. All magnetic geometry parameters are held fixed at the

Group	Varied input	Training range	Outside range
ood_RLTS_1	R/L_{Te}	1.5–3.5	3.5–5.0
ood_RLTS_2	R/L_{Ti}	1.5–3.5	3.5–5.0
ood_RLN	R/L_n	1.0–3.0	3.0–4.5
ood_TAUS_2	T_i/T_e	0.5–1.5	1.5–2.5
ood_XNUE	ν_{ei}	0.01–1.0	1.0–5.0
ood_Q_LOC	q	1.5–2.5	2.5–4.0
ood_KAPPA_LOC	κ	1.0–2.0	2.0–2.5
ood_DRMAJDX_LOC	Δ'	−0.5–0.0	−0.75–−0.5
ood_RMIN_LOC	r/a	0.25–0.75	0.75–0.90

Table 1. *The nine* scan_db_outside *single-parameter out-of-distribution sweeps. Each group varies only the listed input beyond its* scan_db *training range, holding the remaining ten inputs inside their nominal ranges.*

exact DIII-D equilibrium values for that radius. As of this checkpoint, TGLF had completed 52 of the 1,000 generated cases; the remaining simulations were still running at submission time, so this database is not yet used for training.

1.4 Summary of Findings and Report Roadmap

On the 43 starter scans, greater model flexibility did not consistently improve prediction, and low aggregate error coexisted with wrong flux signs. On the larger scan_db database, a multilayer perceptron (MLP) had the lowest error and the fastest prediction of the models we compared, and the accuracy ensemble reduced test root mean squared error (RMSE) by 59.6%–77.2% relative to a linear baseline. The quantile ensemble's 90% intervals covered 94–98% of test cases and were widest where training data was sparse. Error rose on inputs outside the training range, and at the DIII-D reference case the surrogate predicted higher fluxes than TGLF did.

Section 2 describes the data preparation and each model (Table 2). Section 3 reports accuracy and Section 4 reports uncertainty and outside-range behavior. Section 5 covers limitations and the planned DIII-D study.

2 Methods

Several models appear in this report. Table 2 gives the name used for each, what it is, and where it is described and reported. The accuracy, quantile, and 15-network ensembles are separate experiments with separate training runs, so their results should not be mixed.

Name	What it is	Described	Reported
Starter-scan models	Mean baseline, linear, spline, quadratic, and random forest fits on the 43 starter cases	§2.3	§3.1
Model-family comparison	Random forest, extreme gradient boosting (XGBoost), and a single MLP, each on raw and log targets, trained on `scan_db`	§2.4	§3.2
Accuracy ensemble	Five MLPs with different seeds, averaged, predicting one value per flux; includes the activation search	§2.5	§3.3, §3.4
Quantile ensemble	Five MLPs that each predict the 5th, 50th, and 95th percentile of every flux (pinball loss), giving a 90% interval	§2.7	§4.1 to §4.4
15-network ensemble	Fifteen MLPs whose intervals come from disagreement between members	§2.8	§4.5

Table 2. *Models in this report, the name used for each, and where it is described and reported.*

2.1 Data Preparation, Input Features, and Target Transforms

We parsed one row per case, checking numeric values, species identity, and flux-file length before assigning output labels. In `scan_db`, we fixed a 70%/15%/15% training, validation, and test split using random seed 42. However, further analysis of our training data revealed four cases with negative ion heat flux, a physically impossible trait in our tokamak simulation. Since all four cases had been assigned to training, and these do not reflect outlier values, rather real, underlying issues with the TGLF data generation, the rows were removed. However, the remaining splits were reserved, leaving 3,496 training, 750 validation, and 750 test cases. Negative particle flux alone was not an exclusion criterion.

The larger-dataset models use 11 inputs: separate temperature gradients, a shared density gradient, the ion-to-electron temperature ratio, safety factor, elongation, flux-surface shift, normalized minor radius, ion mass, logarithmic collision frequency, and magnetic shear. We removed constant and duplicate coordinates. Magnetic shear was reconstructed from the TGLF inputs as

$$\hat{s} \equiv \frac{r}{q}\frac{dq}{dr} = \frac{q'(r/a)^2}{q^2},$$

where $q = \texttt{Q_LOC}$ is the safety factor, $q' = \texttt{Q_PRIME_LOC}$ is the TGLF safety-factor gradient parameter, and $r/a = \texttt{RMIN_LOC}$ is the normalized minor radius. The conversion assumes $q \neq 0$. Identifiers and other flux outputs were excluded from the predictors.

The positive heat targets span several orders of magnitude. We modeled $\log_{10} Q_i$ and $\log_{10} Q_e$, while retaining signed Γ_e on its original scale. Inputs and transformed targets were standardized using training rows only. Heat predictions were converted back to flux units before evaluation. This transformation changes the fitting objective; improved fit in logarithmic space need not imply lower error in original units.

2.2 Exploratory Data Analysis on Starter Scans

Exploratory data analysis of the starter scans reveals insights that can guide future model development and analysis. We first consider the outputs themselves (Figure 1):

1. The heat fluxes span more than an order of magnitude. Across the 43 cases, ion heat flux ranges from 4.4 to 122.8 (about 28-fold) and electron heat flux from 2.0 to 123.2 (about 63-fold). On a logarithmic axis these values are spread more evenly, which is why we model the heat fluxes in logarithmic space.
2. Particle flux behaves differently. It ranges from −0.39 to 44.5, and two cases have negative (inward) flux, so a logarithm cannot be applied. Most cases are small, with 32 of 43 below 4 and a long tail of larger values. We therefore keep particle flux signed on its original scale.

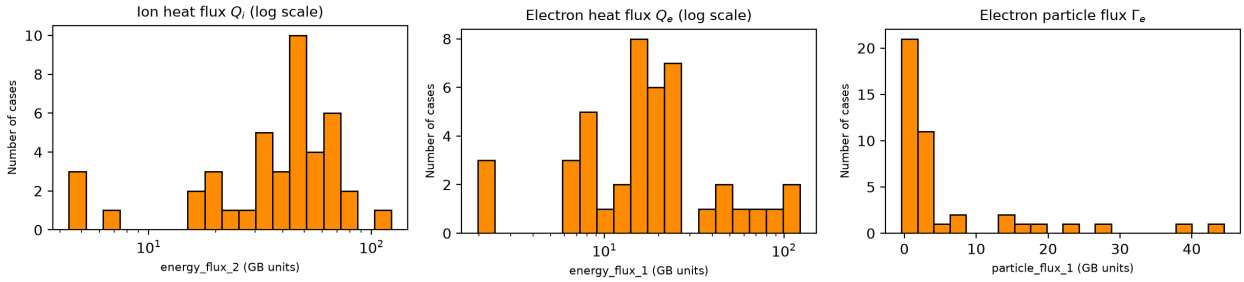


Figure 1. *Distributions of ion heat flux, electron heat flux, and electron particle flux across the 43 starter cases, in gyro-Bohm units. The heat flux axes are logarithmic; the particle flux axis is linear and includes two negative values.*

We then analyze the inputs using the correlation heatmap in Figure 2:

1. Few inputs correlate strongly with the fluxes. The highest correlation with ion heat flux Q_i is 0.55 (with the electron temperature gradient), so a single input explains only about 30% of its variance in a straight-line fit. A linear model on individual inputs cannot approximate all of the fluxes well.
2. Some input–output pairs are strongly correlated, such as the electron density gradient with electron particle flux (0.81) and with electron heat flux (0.75). This points to a comparatively simple dependence for those pairs. The starter scans change some inputs together, so these correlations may also reflect that coupling and should not be read as isolated effects.
3. The outputs move together: Γ_e and Q_e correlate at 0.98, and Q_i correlates with Q_e at 0.80 and with Γ_e at 0.66. This supports predicting the three fluxes with a shared multi-output model.

Finally, we examine how the outputs respond to individual inputs. Figure 3 illustrates why we consider nonlinear models. Beyond the fact that we are trying to approximate complex physical interactions inside tokamaks, we notice the following:

1. The increase in output fluxes differs across input variables, suggesting nonlinear relationships.

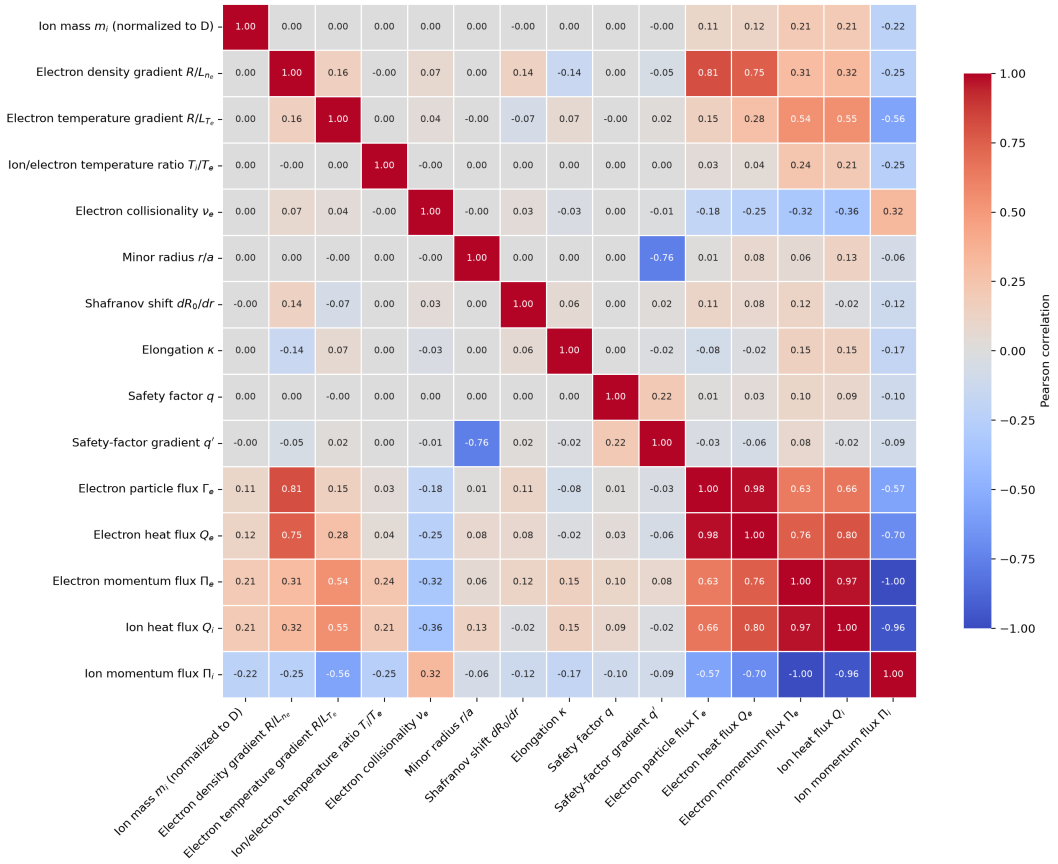


Figure 2. *Pearson correlation between the inputs that vary across the 43 starter cases and the output fluxes. Two duplicate columns (*`RLNS_2`*,* `RLTS_2`*) and* `particle_flux_2` *are omitted because they equal* `RLNS_1`*,* `RLTS_1`*, and* `particle_flux_1`*. Momentum fluxes are shown but are not modeling targets.*

2. The faded points in many of the plots show that varying other inputs simultaneously changes the output flux substantially.
3. In some cases, such as the ion temperature gradient scan, the flux relative to deuterium varies nonlinearly across ion species.

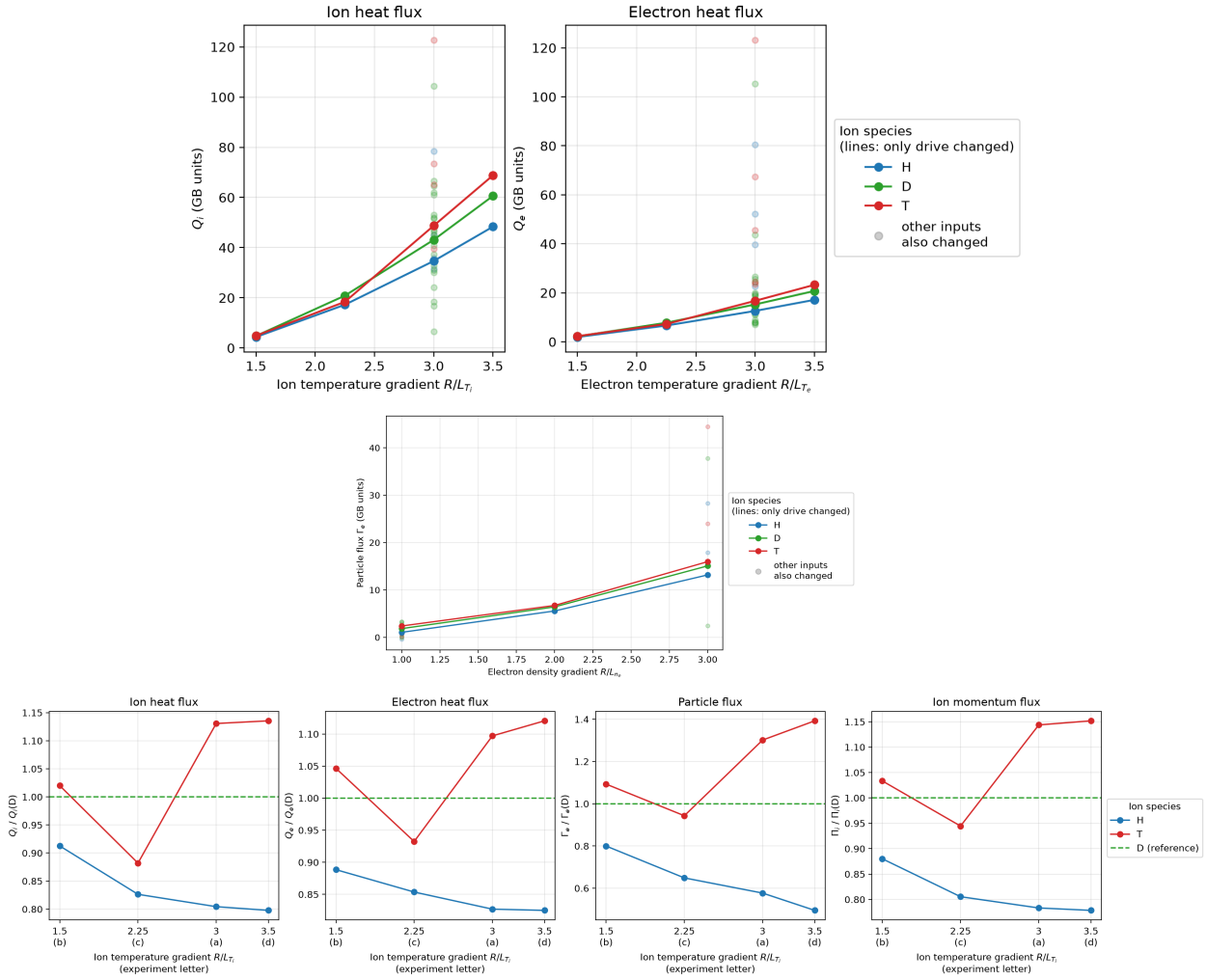


Figure 3. *Starter-scan responses in gyro-Bohm units. Top: ion and electron heat flux against the matching temperature gradient. Middle: particle flux against density gradient. In both, lines join cases where only the shown input (and the isotope) changed, and faded points are cases where other inputs also changed. Bottom: hydrogen and tritium flux divided by deuterium flux across the ion temperature-gradient scan, labeled by experiment letter. These are one-at-a-time scans of three to four points on 43 cases, so they illustrate input interactions rather than establish the shape of any response.*

2.3 Exploratory Regression and Forest Models on the Starter Scans

The initial comparison used ten physical input coordinates on the 43 starter cases, representing the tied temperature gradients with one coordinate. We compared a training-mean baseline, linear regression variants, additive splines, quadratic terms with interactions, and a random forest. The variants tested logarithmic collision frequency and, for heat fluxes,

logarithmic targets. The forest used 100 trees, maximum depth four, and a minimum leaf size of three.

For outer evaluation, we held out one entire letter group at a time, keeping related isotope cases together. Inner five-fold validation, also grouped by letter, selected regression ridge penalties from $\{0.1, 1, 10, 100\}$ using case-level mean absolute error (MAE). Preprocessing was fitted within the training folds. The forest and mean baseline had fixed settings. We report outer MAE averaged first within each letter and then across letters, giving each group equal weight. Model families were compared using these same outer scores, so this is an exploratory screen rather than an untouched final test. The results are in Section 3.1 (Table 3).

2.4 Model-Family Comparison Setup on `scan_db`

Before moving forward with the accuracy ensemble, we compared the efficiency and accuracy of random forest, XGBoost, and single MLP models. To represent the accuracy of each model fairly, all three architectures were trained on both raw and log-transformed flux outputs, and all were tuned with the same cross-validation function through grid search. A fixed ridge regression serves as the baseline. We report normalized root mean squared error (nRMSE), which is the RMSE of each flux in units of that flux's training-set standard deviation, averaged over the three fluxes. All reported nRMSE and R^2 values are in raw flux units, so raw-target and log-target models can be compared. The timing, learning-curve, and SAT3 results are in Section 3.2.

2.5 Five-Network Ensemble and Activation Search

Building off of the promising results of the single MLP in the model-family comparison (Section 3.2), we compared the accuracy ensemble, five MLPs, with two baselines on `scan_db`. The mean baseline predicts each training target's arithmetic mean in flux units. The linear baseline uses standardized inputs and the same transformed heat and signed particle targets as the networks, without regularization or interaction terms. These baselines differ from the grouped ridge models in the starter-data experiment.

Each network has layer widths 11–64–64–8–3 and a linear output layer. The five members use seeds 42–46. Training uses adaptive moment estimation with decoupled weight decay (AdamW), batch size 256, and weight decay 10^{-4}, with 30 warmup epochs at learning rate 0.001 and up to 120 further epochs at 0.003. Validation selects the saved checkpoint, with early stopping after 25 epochs without improvement. Each member's heat outputs are inverted before taking the equal-weight average of all five predictions.

We tested four combinations of the Gaussian error linear unit (GELU), rectified linear unit (ReLU), and hyperbolic tangent (tanh) activations. The first activation was applied to both 64-unit layers, and the second to the 8-unit layer: GELU with tanh, ReLU with tanh, GELU with ReLU, and ReLU throughout. Across combinations, we matched initial weights and minibatch order within each seed and held the architecture, data, scalers, and training set-

tings fixed. Using each target's training standard deviation (SD), we computed the selection score

$$S = \frac{1}{3} \sum_{j \in \{Q_i, Q_e, \Gamma_e\}} \frac{\mathrm{RMSE}_{\mathrm{val},j}}{\mathrm{SD}_{\mathrm{train},j}}, \tag{1}$$

with both quantities measured in original flux units. This prevents a large-scale target from dominating selection. The learning rate came from an earlier GELU/tanh search; the alternatives were not separately tuned. GELU/ReLU was selected, and its five checkpoints were retained unchanged.

This search was limited to four activation combinations on one architecture and one data split, using five matched seeds and a shared training schedule. The improvement in aggregate validation error is promising, although the selected configuration increased electron-heat error relative to the control. These findings motivate further tests across architectures and separately tuned training settings rather than a final choice of activation functions. The results are in Section 3.3 (Tables 5 and 6).

2.6 Evaluation Protocol, Learning Curves, and Test-Set Caveats

We report RMSE and R^2 in addition to the selection score. Learning curves use nested training subsets of 250, 500, 1,000, 2,000, and 3,496 cases, defined by a fixed permutation with seed 2026. Scalers were refitted on each subset, with the same validation set for checkpoint selection. We also evaluated the selected ensemble on all 43 starter cases and checked the directions of temperature-gradient and isotope trends. These cases were already used for exploration, but did not select the activation configuration.

The 750-case random test set had been inspected in earlier work. Consequently, the reported comparisons are exploratory and do not constitute a new, untouched assessment. The learning curves also use only one nested sequence. With a fixed maximum epoch budget, larger subsets receive more minibatch updates per epoch, so the curves do not isolate data volume from optimization effort. The learning-curve results are in Section 3.4 (Figure 7), and the starter-set trend checks are in Section 3.5.

2.7 Prediction Intervals I: Quantile-Regression Ensemble

The main interval method is the quantile ensemble. Instead of predicting one value per flux, each network predicts the 5th, 50th, and 95th percentiles of that flux, so its output layer has nine units (three fluxes times three quantiles). The median is the point prediction, and the outer two quantiles form a 90% interval. For quantile q and error $e = y - \hat{y}$, the pinball loss is

$$L_q(e) = \max\{\, q\, e,\ (q-1)\, e \,\}. \tag{2}$$

Under-prediction costs q per unit of error and over-prediction costs $1-q$. At $q = 0.05$, over-prediction is far more expensive, so the output settles near a value that only 5% of the targets fall below. At $q = 0.95$ the penalty is reversed and the output settles near the 95th

percentile. The further q is from 0.5, the more lopsided the penalty. The raw outputs can cross, so the three quantiles are sorted before use. Lower pinball loss is better.

Each network has layer widths 11–64–64–8–9 with GELU, GELU, and ReLU activations. Five members differ only in their random seed, and their outputs are averaged. The 11 inputs, the data split, and the $\log_{10}$ transformation of the heat fluxes follow Section 2.5, and Γ_e stays signed on its original scale. The training schedule also follows that section, with 30 warmup epochs at learning rate 0.001 and 120 epochs at 0.003, but this model uses Adam with full-batch updates, a fixed 150 epochs, and no checkpoint selection or early stopping. The validation and test sets play no part in training. It is a separate experiment from the accuracy ensemble.

We report R^2 and nRMSE of the median, and coverage, the fraction of true values inside the interval (nominal 0.90). Interval width is ln(upper/lower) for the heat fluxes, which is a relative width. For Γ_e it is upper − lower in gyro-Bohm units, an absolute width that grows with the flux, and upper − lower divided by the median (where the median is positive) when we need a relative width. We evaluated the ensemble without retraining on nine groups of 30 outside-range cases, each pushing one input beyond the scan, and on a DIII-D reference case at $r/a = 0.5$. For the DIII-D case, 10,000 random input sets were drawn uniformly within the input error bars and passed through the ensemble. These draws have no TGLF counterparts. We compared the ensemble separately with 300 TGLF runs at inputs sampled within the same error bars. The results are in Sections 4.1 to 4.4.

2.8 Prediction Intervals II: Member Disagreement

For this uncertainty experiment, we expanded the ensemble from 5 to 15 neural networks to bring it closer in size to the 32-network ensemble that Meneghini et al. (2017, Section 3.1) used for their TGLF surrogate. We estimated uncertainty from disagreement among the 15 members. For each flux, prediction intervals were centered on the ensemble mean. Their half-width was a calibrated multiplier times the larger of the member sample standard deviation and 1% of the training-target standard deviation. Multipliers were fitted on the 750 validation cases for nominal 90% coverage, then fixed for subsequent checks. Because validation had already supported model selection, coverage is an empirical assessment rather than a formal guarantee. These results are reported separately from the five-network activation comparison and do not measure the effect of increasing ensemble size. The results are in Section 4.5.

2.9 Gradient-Boosted Histogram-Tree Surrogate and Bootstrap Prediction Intervals

As an independent, non-neural-network surrogate family on `scan_db`, we trained `scikit-learn`'s `HistGradientBoostingRegressor` for each of Q_i, Q_e, and Γ_e. Gradient boosting builds an additive model $F_M(\mathbf{x}) = \sum_{m=1}^{M} \eta\, h_m(\mathbf{x})$ by fitting each successive weak learner h_m to the negative gradient of the loss with respect to the current ensemble output (Friedman 2001).

For squared-error loss, this pseudo-residual at boosting round m is the ordinary residual, $r_i^{(m)} = y_i - F_{m-1}(\mathbf{x}_i)$, and h_m is a regression tree fit to $\{(\mathbf{x}_i, r_i^{(m)})\}$. The learning rate η shrinks each tree's contribution before it is added to the ensemble. The *histogram* variant discretizes each continuous feature into at most 256 bins before growing each tree, accumulating per-bin gradient and Hessian sums so that the best split at a node is found by scanning bin boundaries rather than every unique training value, following the binning strategy introduced for gradient boosting by Ke et al. (2017). This reduces split-finding cost per node from $O(n \log n)$ to $O(n_{\text{bins}})$ and lets each of our three single-target models fit on the 3,497 training cases in 1–2 seconds. Each model used the 11 inputs from Section 2 (unmodified, without the engineered magnetic-shear substitution) and $\log_{10}(\texttt{XNUE})$, predicting the raw, untransformed flux. Hyperparameters (learning rate, maximum leaf nodes per tree, and minimum samples per leaf) were selected by a grid search scored on the validation partition, with early stopping on a held-out fraction of the training rounds.

To attach a prediction interval to this surrogate family, we trained a 10-member bootstrap ensemble per target: for $m = 1, \ldots, 10$, a histogram-gradient-boosting model with fixed hyperparameters was fit to a resample (with replacement, same size as the training set) of the 3,497 training cases, using a distinct seed for each member. For an input vector $\mathbf{x}$, the ensemble mean $\bar{y}(\mathbf{x}) = \frac{1}{M}\sum_{m=1}^{M} \hat{y}^{(m)}(\mathbf{x})$ is the point prediction, and the sample standard deviation $s(\mathbf{x}) = \sqrt{\frac{1}{M-1}\sum_{m=1}^{M}(\hat{y}^{(m)}(\mathbf{x}) - \bar{y}(\mathbf{x}))^2}$ quantifies epistemic spread across the bootstrap resamples. Because this raw spread need not give honest coverage, we calibrated it on the validation partition: for each validation case i we computed the normalized residual $z_i = |y_i - \bar{y}(\mathbf{x}_i)| / \max(s(\mathbf{x}_i), 10^{-4})$, and took $c_{0.90}$, the empirical 90th percentile of $\{z_i\}$, as a per-target scaling factor. The resulting calibrated 90% prediction interval for a new query $\mathbf{x}$ is $[\bar{y}(\mathbf{x}) - c_{0.90}\, s(\mathbf{x}),\ \bar{y}(\mathbf{x}) + c_{0.90}\, s(\mathbf{x})]$. The fitted multipliers were $c_{0.90} = 2.616$ for Q_i, 2.721 for Q_e, and 2.631 for Γ_e, each several times larger than the nominal Gaussian value of 1.645, meaning the raw bootstrap spread alone would have substantially under-covered the validation targets. This is a separate experiment from the quantile and 15-network ensembles in Sections 2.7 and 2.8: it trains different model families on an independently drawn 70%/15%/15% split of the same `scan_db` data (3,497 training, 749 validation, and 750 test cases after removing the same four negative-Q_i cases), so its coverage and error figures should not be pooled with those of the neural-network ensembles. Results are in Sections 3.6 and 4.6.

3 Accuracy Results

3.1 Starter-Scan Comparison: Flexibility and Failure Modes

These results use the setup in Section 2.3. Table 3 shows that greater flexibility did not consistently improve the starter-data predictions. Replacing raw collision frequency with its logarithm improved linear-model MAE for all three targets. The fixed forest performed worse than that transformed linear model on each target. The logarithmic heat model had the lowest electron-heat MAE, while quadratic terms had the lowest ion-heat MAE.

Model	Q_i	Q_e	Γ_e
Training mean	17.022	14.830	6.127
Raw linear	9.570	10.215	3.520
Log-collision linear	8.295	7.695	2.725
Additive splines	17.528	11.475	3.911
Quadratic + interactions	7.965	7.253	3.210
Random forest	16.302	11.267	3.484
Log-heat linear	10.213	5.306	—

Table 3. *Exploratory starter-data comparison: letter-balanced MAE in gyro-Bohm units across 26 held-out letter groups. Lower is better. These scores use a different dataset and procedure from Table 5. The logarithmic heat model was not applied to particle flux.*

The lowest MAE did not ensure suitable behavior in every case. The quadratic ion-heat model predicted −1.34 and −20.78 for cases with positive TGLF values of 4.90 and 6.49, respectively. This motivates examining prediction signs and individual failures when selecting a model. The comparison concerns 43 structured cases and the settings tested here; it does not establish that forests or splines are generally unsuitable for TGLF.

3.2 Model-Family Comparison on `scan_db`: Speed, Learning Curves, and SAT3 Transfer

The comparison set up in Section 2.4 shows the following:

1. The random forest had the shortest training time per fit (0.43 s on raw targets and 0.92 s on log targets, compared with 11.1 s and 12.7 s for the MLP), but it had the longest prediction time of all the models (Figure 4). Predicting one case at a time took about 40 ms on raw targets and 88 ms on log targets for the forest, compared with about 0.22 ms and 0.19 ms for the MLP. The neural network showed the opposite pattern, with a long training time but a short prediction time. Because we need a fast-inference surrogate, the MLP better fits our needs.
2. The learning curves (Figure 5) show that the error of all three models decreased as the training set grew, especially compared with the ridge baseline. The MLP had the lowest validation nRMSE at every training size, clearly outperforming XGBoost and the random forest. With all 3,496 training cases, the MLP reached 0.233 on raw targets and 0.215 on log targets, compared with 0.395 and 0.355 for XGBoost and 0.510 and 0.538 for the random forest.
3. After training on `scan_db`, we tested the models on the held-out SAT3 cases (Figure 6). The MLP trained on log-transformed targets performed best by far, with a SAT3 nRMSE of 0.035 and R^2 of 0.987, compared with nRMSE of 0.117 for XGBoost and 0.121 for the random forest on log targets. XGBoost and the random forest were also clearly worse than the MLP on the `scan_db` test set. The R^2 values tell the same story: the MLP is best by far, while the ridge baseline and the random forest on raw

targets (SAT3 R^2 of −0.55 and −0.07) performed worse than simply predicting the mean.

Overall, the MLP performs best. Every model has a lower nRMSE on SAT3 than on the `scan_db` test set, so the SAT3 values are most useful for ranking the models, not for judging absolute accuracy.

An earlier run of the same comparison, which used the original TGLF inputs before we added the engineered magnetic shear and logarithmic collision frequency inputs, corroborates this ordering (Table 4). The MLP on log-transformed targets again performed best by a wide margin (SAT3 nRMSE of 0.075 and R^2 of 0.951, compared with 0.235 and 0.486 for XGBoost), and the random forest again had a negative SAT3 R^2 (−0.20 on log targets and −0.90 on raw targets). Absolute errors were higher for every model in the earlier run, so we use it to check the ordering of the models, not their absolute accuracy.

		SAT3		`scan_db` test	
Model	Targets	nRMSE	R^2	nRMSE	R^2
MLP	log	0.075	0.951	0.331	0.888
MLP	raw	0.163	0.758	0.364	0.865
XGBoost	log	0.235	0.486	0.382	0.850
XGBoost	raw	0.299	0.239	0.400	0.836
Random forest	log	0.367	−0.196	0.554	0.685
Random forest	raw	0.469	−0.899	0.551	0.688

Table 4. *Earlier run of the model-family comparison with the original TGLF inputs, scored in raw flux units. Models were trained on* `scan_db` *only. This run did not include the engineered shear and logarithmic collision frequency inputs, so absolute values differ from Figure 6.*

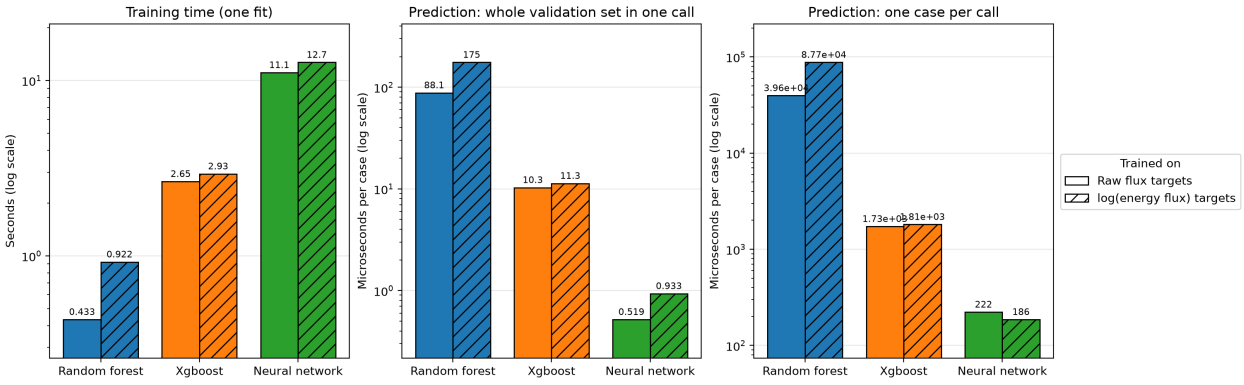


Figure 4. *Training time (left) and prediction time per case (middle: whole validation set in one call; right: one case per call) for each model. Prediction times are medians over repeated calls. Hatched bars are models trained on log-transformed heat flux targets. Axes are logarithmic.*

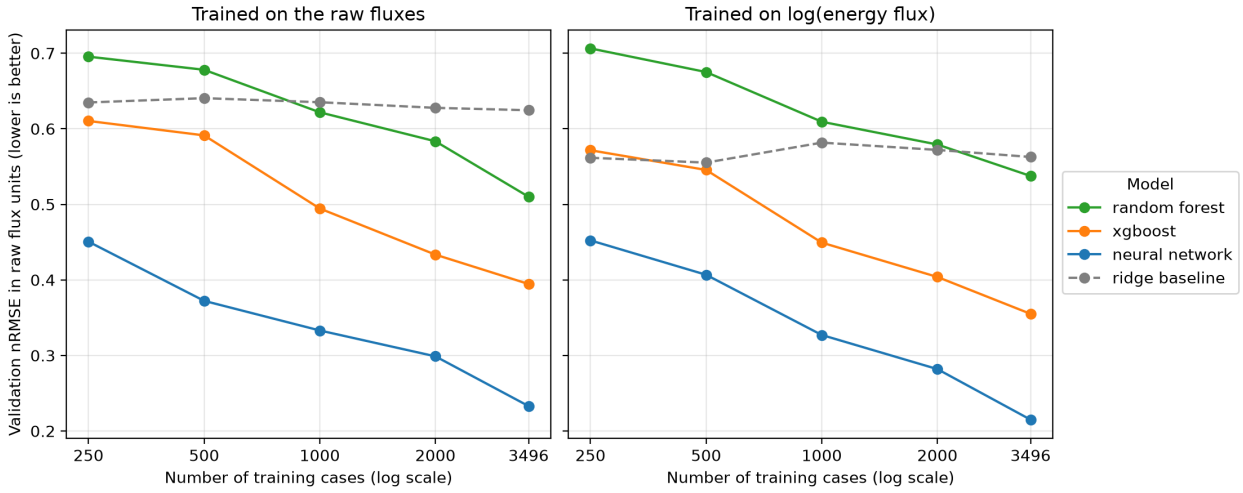


Figure 5. *Validation nRMSE against training set size for each model at its tuned hyperparameters. Left: models trained on raw fluxes. Right: models trained on log-transformed heat fluxes. The dashed line is the ridge baseline.*

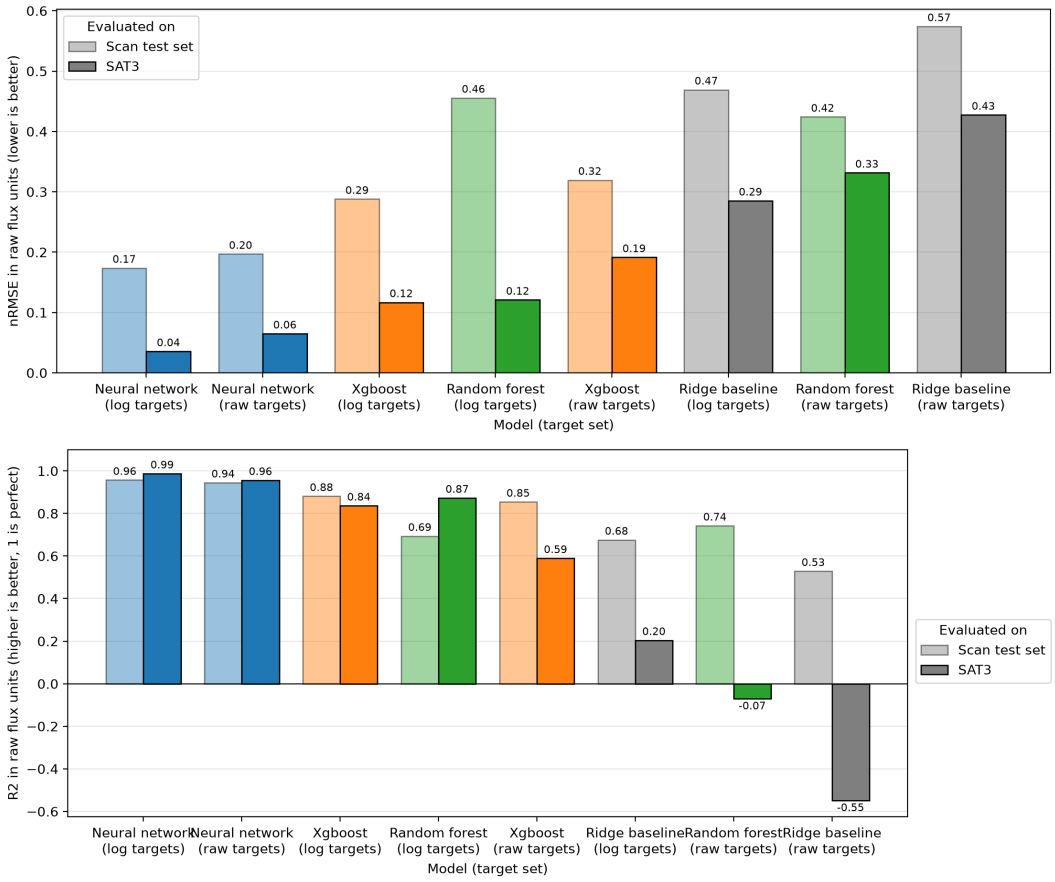


Figure 6. *nRMSE (top) and R^2 (bottom) on the* `scan_db` *test set (faded bars) and on the 43 SAT3 cases (solid bars) for each model and target set. Models were trained on* `scan_db` *only, and nothing was refit for SAT3. R^2 below zero means worse than predicting the mean.*

3.3 Ensemble Accuracy Against Baselines and Activation Choice

On the larger database, the accuracy ensemble outperformed the mean and linear baselines on all three outputs (Table 5; setup in Section 2.5). Relative to linear regression, test RMSE decreased by 59.6% for ion heat, 61.8% for electron heat, and 77.2% for particle flux. This does not establish superiority over tree models evaluated elsewhere, or isolate the benefit of averaging compared with an otherwise matched single network.

	RMSE			R^2		
Model	Q_i	Q_e	Γ_e	Q_i	Q_e	Γ_e
Mean	76.53	60.20	20.93	−0.009	−0.010	−0.011
Linear	38.43	29.48	14.43	0.746	0.758	0.519
Five-network ensemble	15.53	11.27	3.29	0.959	0.965	0.975

Table 5. *Performance on the same 750 random test cases from* `scan_db`*. RMSE is in gyro-Bohm units. The test set was previously inspected; these are exploratory results for the selected GELU/ReLU model.*

The controlled activation comparison selected GELU/ReLU, with a mean normalized validation RMSE of 0.1806, compared with 0.1887 for GELU/tanh (Table 6). The 4.3% aggregate improvement came from ion heat and particle flux; electron-heat error increased. Replacing all hidden activations with ReLU increased the aggregate error by 11.5% relative to GELU/tanh. Thus, neither a uniform activation replacement nor a single aggregate score captures every target's response. These results support the selected configuration under the fixed training setup, rather than a universal ordering of activation functions.

Activations	S	Q_i RMSE	Q_e RMSE	Γ_e RMSE
GELU / ReLU	0.1806	15.80	13.16	4.61
GELU / tanh	0.1887	17.62	12.76	4.83
ReLU / tanh	0.1935	18.15	12.93	4.99
ReLU / ReLU	0.2105	18.67	15.62	5.19

Table 6. *Activation comparison on 750 validation cases. The first activation applies to both 64-unit layers and the second to the 8-unit layer. RMSE values are in gyro-Bohm units; S is defined in Eq. 1.*

3.4 Learning Curves and Computational Cost

The ensemble's errors decreased as the training set grew (Figure 7; protocol in Section 2.6). Mean normalized test RMSE decreased by 50.2% between 250 and 3,496 training cases, compared with 2.9% for linear regression. The ensemble improved by a further 11.1% between 2,000 and 3,496 cases. This run provides evidence that the model still benefited from additional training examples over the tested range. It does not establish the amount of data needed to reach a specified accuracy.

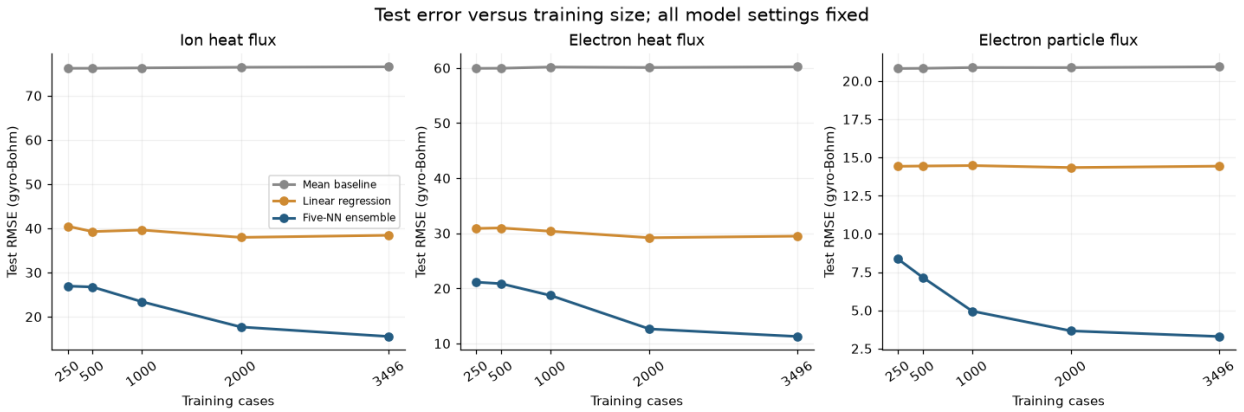


Figure 7. *Learning curves for the selected GELU/ReLU five-network ensemble and both baselines. Each point uses the same previously inspected 750-case test set. Subsets form one nested sequence, with no resampling uncertainty bands. Settings are fixed, but larger subsets entail more minibatch updates per epoch.*

Training all five networks on a central processing unit (CPU) on UC San Diego's Data Science/Machine Learning Platform (DSMLP) took 24.01 s. Median prediction latency was 0.814 ms for one case, while a batch of 750 averaged 4.51 μs per case. Training used four threads and prediction used one. Prediction timings used seven repetitions after warmup and included scaling, inverse transformations, and averaging, but excluded model loading, file access, and feature construction. The supplied TGLF logs do not provide elapsed runtimes, so these measurements do not support a measured TGLF-to-surrogate speedup ratio.

3.5 Starter-Set Trend and Sign Checks

On the 43 starter cases, the selected ensemble achieved R^2 values of 0.9702, 0.9973, and 0.9876 for Q_i, Q_e, and Γ_e, respectively. Nevertheless, it matched only 25 of 27 temperature-gradient directions and 45 of 48 isotope directions, and predicted the wrong particle-flux sign in two cases. The validation-selected activation also improved starter-set R^2 for electron heat relative to GELU/tanh while reducing it for ion heat and particle flux. These checks show why an overall score should be read alongside local trends and failure cases.

3.6 Gradient-Boosted Surrogate on `scan_db`: Accuracy and Speed

Table 7 reports the tuned `HistGradientBoostingRegressor` surrogate from Section 2.9 on its own 750-case `scan_db` test partition, alongside the mean and ridge baselines fit on the same split. We report this as a separate table rather than adding rows to Table 3 or Table 5: Table 3 scores a different dataset (the 43 starter cases) and metric (letter-grouped MAE), and Table 5 is captioned as the same 750 test cases used to select the five-network ensemble's activation function, whereas this surrogate used its own independently drawn split of `scan_db`, so its 750 test cases are not the same rows. The gradient-boosted surrogate reduced test RMSE relative to its ridge baseline by 43.5% for Q_i, 38.6% for Q_e, and 53.6% for

Γ_e, broadly consistent in direction with the larger reductions reported for the five-network ensemble in Section 3.3, though the two are not evaluated on the same split and should not be compared as if they were.

	RMSE			R^2		
Model	Q_i	Q_e	Γ_e	Q_i	Q_e	Γ_e
Mean	89.64	62.41	22.81	−0.000	−0.002	−0.002
Ridge	62.31	42.00	15.74	0.517	0.546	0.523
GB surrogate	35.20	25.80	7.30	0.846	0.829	0.897

Table 7. *Test-set performance of the tuned* `HistGradientBoostingRegressor` *surrogate (learning rate 0.08, 31 maximum leaf nodes, minimum leaf size 10) and its baselines on 750* `scan_db` *test cases, a different split from Table 5. RMSE is in gyro-Bohm units.*

Training each single-target model (300 boosting rounds with early stopping, 3,497 training cases) took 1.18 s for Q_i, 1.69 s for Q_e, and 1.26 s for Γ_e on a single CPU core. Per-case prediction latency, measured by repeating a 750-case batch call 100 times and dividing by the total number of evaluations, was 12 μs for Q_i, 15 μs for Q_e, and 12 μs for Γ_e — roughly five orders of magnitude faster than the 3-second per-case TGLF runtime assumed for this comparison, giving speedup factors of about 2.4×10^5, 2.0×10^5, and 2.5×10^5, respectively. These timings measure batched, amortized per-case cost rather than isolated single-case calls, so they are not directly comparable to the single-call latency reported for the five-network ensemble in Section 3.4.

4 Uncertainty and Out-of-Range Results

This section evaluates the prediction intervals from the two methods in Sections 2.7 and 2.8, in the same order. Sections 4.1 to 4.4 report the quantile ensemble: its coverage, what its interval width tracks, its behavior on inputs outside the training range, and the DIII-D reference case. Section 4.5 reports the 15-network ensemble. All fluxes are in gyro-Bohm units, and all comparisons are against TGLF, not experimental measurements.

4.1 Interval Coverage and Calibration

Table 8 gives the accuracy of the median prediction and the coverage of the 5–95% interval. On the test set the median reached R^2 of 0.876, 0.804, and 0.814 for Q_i, Q_e, and Γ_e, and coverage was 94.8%, 93.9%, and 98.0%. Coverage is above the nominal 90%, so the intervals are somewhat conservative, most of all for particle flux. The quantile intervals needed no separate calibration step.

Figure 8 shows the median and interval against the true flux for 250 randomly chosen test cases. The widest intervals sit at the extremes of the flux range, and most of the ion-heat misses are at low true flux. During training, the mean interval width on the training set

Split	Target	R^2 (median)	nRMSE (median)	Coverage
Validation	Q_i	0.879	0.347	0.956
Validation	Q_e	0.729	0.477	0.951
Validation	Γ_e	0.817	0.408	0.980
Test	Q_i	0.876	0.304	0.948
Test	Q_e	0.804	0.365	0.939
Test	Γ_e	0.814	0.354	0.980

Table 8. *Accuracy of the median and coverage of the 5–95% interval for the quantile ensemble. Nominal coverage is 0.90. nRMSE is RMSE divided by the training-set standard deviation of that flux.*

rose sharply after the warmup epochs, peaked near epoch 50, and then fell while the loss kept decreasing. It was still falling at epoch 150, so the interval widths depend on how long the networks train.

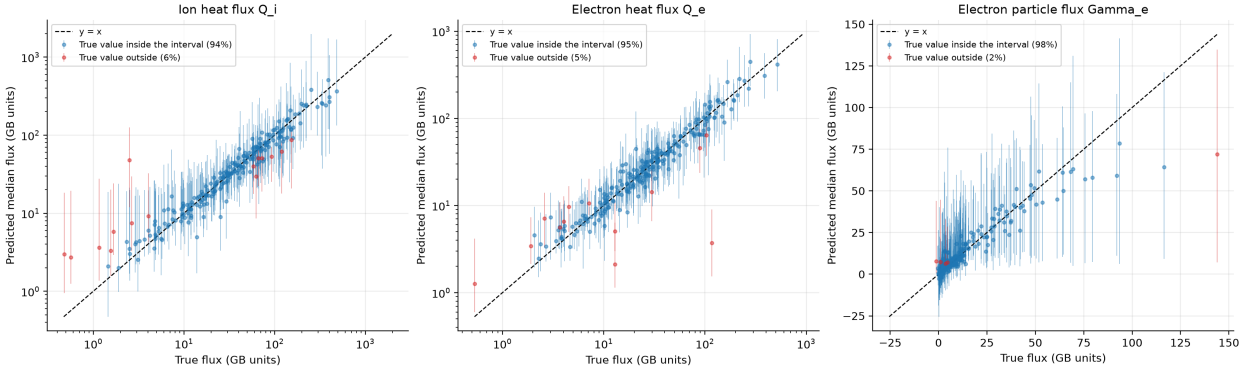


Figure 8. *Predicted median and 5–95% interval against the true flux for 250 random test cases. Blue intervals contain the true value and red ones do not. The dashed line is $y = x$. Heat-flux axes are logarithmic.*

4.2 Interval Width as an Error Indicator

Wider intervals go with larger errors (Figure 9). On the test set the Spearman correlation between band width and absolute error is 0.46 for Q_i, 0.41 for Q_e, and 0.68 for Γ_e. The scatter is large, so width is a rough warning and not a precise error estimate. For Γ_e, width and error are both in raw units and both grow with the flux, so part of its correlation is a scale effect.

Figure 10 shows how width depends on the true flux. For Q_i the pattern is a U: intervals are widest at very low flux (below about 3) and very high flux (above about 100), and narrowest between about 20 and 80. These are also the flux ranges with the least training data, which is the main evidence that interval width follows how much data the model has seen. For Γ_e the absolute width increases with flux, but this is mostly scale, because the width stays in raw units. At very low flux the Γ_e widths also sit above their minimum, which is the other arm of the same U. Q_e shows only a weak relationship.

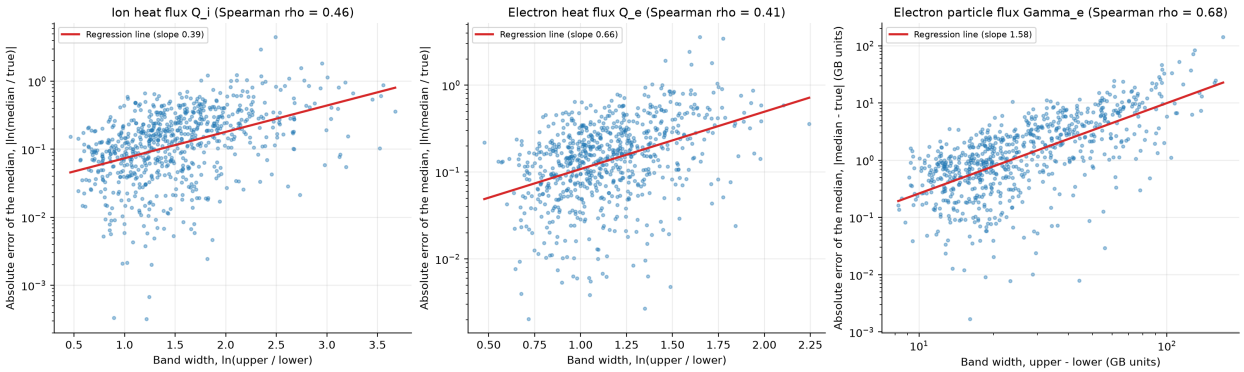


Figure 9. *Absolute error of the median against band width on the test set. The red line is a least-squares fit of the logarithm of the error on the width. Both axes of the* Γ_e *panel are in raw gyro-Bohm units.*

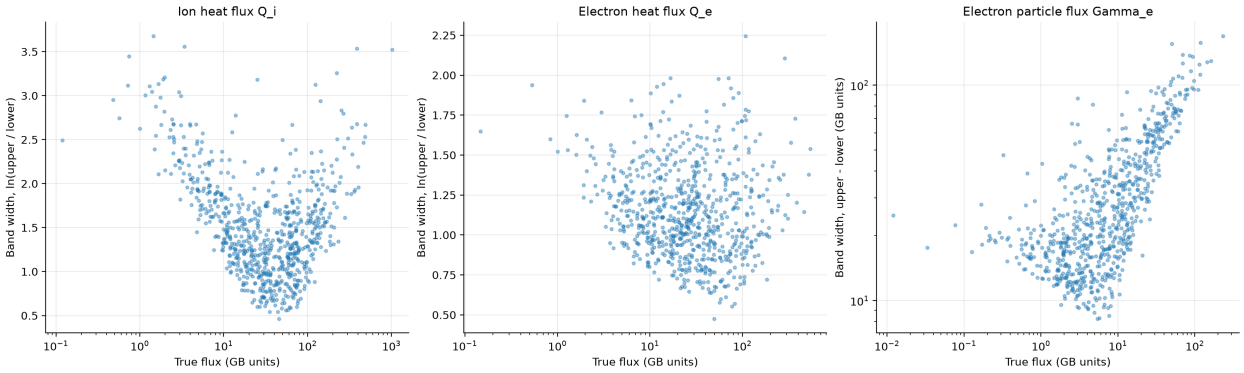


Figure 10. *Band width against the true flux on the test set. The heat-flux widths are* $\ln(\mathit{upper}/\mathit{lower})$*. The* Γ_e *width is upper − lower in gyro-Bohm units, so it grows with the flux.*

Figure 11 tests the data-density explanation directly. We cut each flux range into bins and counted the training cases in each bin. Average relative width fell as the bin contained more training cases, with Spearman correlations of −0.97 for Q_i and −0.78 for Q_e. For Γ_e the per-bin correlation is only −0.03, although the average over the third of bins with the fewest cases (about 17) is far above the third with the most (about 3). A few bins with a median near zero inflate the relative width there. This shows where the intervals are wide, but not whether the network has learned the underlying physics.

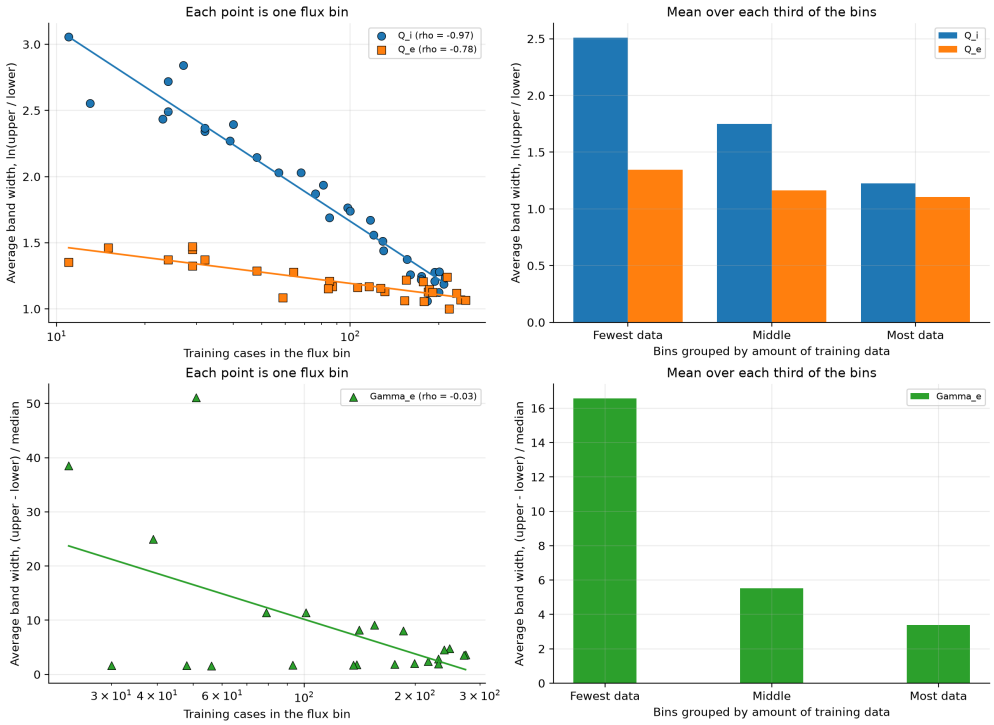


Figure 11. *Average relative band width against the number of training cases in the flux bin (left) and averaged over the third of bins with the fewest, middle, and most training cases (right), on the test set. The top row shows the heat fluxes and the bottom row shows* Γ_e.

The inputs most correlated with relative width differ by flux. For Γ_e it is the density gradient `RLNS_1` (Spearman −0.69 on the test set, so smaller gradients give wider intervals), followed by log collision frequency (0.35). For Q_e it is `RLNS_1` (−0.45) and `RLTS_1` (0.28). For Q_i the correlations are all small: elongation (0.26), `RLTS_2` (−0.27), and `TAUS_2` (−0.25). Inputs also move the flux itself, so part of this reflects flux level, most of all for Γ_e, whose relative width is inflated where the median is near zero.

4.3 Behavior on Out-of-Range Inputs

Error was higher on the outside-range groups than on the ordinary test set in 22 of the 27 group and flux combinations, and up to about ten times higher for Q_e when `RLNS_1` is pushed out of range (Figure 12). Coverage on these groups stayed between 0.80 and 1.00, with 30 cases per group.

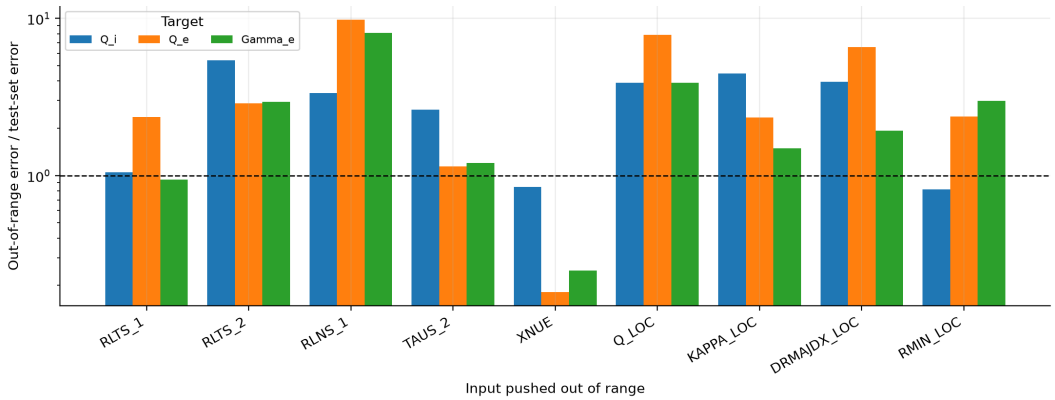


Figure 12. *nRMSE of the median on each outside-range group divided by the nRMSE on the in-range test set. The dashed line at 1 marks equal error. The vertical axis is logarithmic, and each group has 30 cases.*

Interval width followed this only partly. For Q_i and Q_e, width rose with distance beyond the scan range for most inputs (Spearman 0.05 to 0.67, largest for `Q_LOC` and `XNUE`). For Γ_e the correlation was near zero for most inputs. There are also direct counterexamples. For `RLNS_1`, the Q_e nRMSE was 3.59 (against 0.37 in range) and R^2 was −0.52, yet the median relative width was 1.03, slightly below the in-range 1.11. The `XNUE` group went the other way: its nRMSE was lower than in range, but R^2 was lower for all three fluxes (0.51, 0.58, and 0.28 against 0.88, 0.80, and 0.81) and the intervals were wider, for example a median Γ_e relative width of 7.8 against 3.0. A wide interval is therefore a useful warning for most outside-range inputs, but a narrow one does not guarantee accuracy.

4.4 DIII-D Input-Uncertainty Case Study

DIII-D is a tokamak at General Atomics, and r/a is the position inside the plasma as a fraction of its minor radius, where 0 is the center and 1 is the edge. The reference case is one real DIII-D plasma at $r/a = 0.5$, halfway out, expressed as a set of TGLF inputs in the `scan_db` format (the gradients, collision frequency, safety factor, and so on, with all other TGLF settings as in `scan_db`). At these inputs TGLF gives $Q_i = 4.31$, $Q_e = 3.63$, and $\Gamma_e = 0.387$.

Measured plasma profiles are not exact, so each uncertain input carries an error bar: ±10% for the temperature and density gradients, the temperature ratio, and the safety-factor gradient, ±20% for collision frequency, and ±5% for the safety factor. The elongation, Shafranov shift, minor radius, and ion mass are held fixed. This case study asks how much those input errors move the predicted fluxes, and whether the surrogate reproduces that spread. TGLF answers the first question by running 300 input sets sampled within the bars. Input errors of 5–20% spread the TGLF fluxes by ±42%, ±45%, and ±80% of the mean for Q_i, Q_e, and Γ_e, where each figure is half of the 5–95% range as a percentage of the mean. We answer the second question by passing 10,000 random input sets from the same bars through the surrogate, which took 0.041 s in total, and keeping its median prediction. The 10,000 draws are inputs only and have no TGLF counterparts.

Figure 13 shows that the surrogate sits above TGLF. The mean surrogate flux was 5.50 against 4.44 for Q_i, 5.31 against 4.12 for Q_e, and 1.38 against 0.48 for Γ_e. When the surrogate is run on the same 300 input sets as TGLF, the correlation with TGLF is high (0.95, 0.89, and 0.92), so it responds to the inputs in the same direction, but its predictions are above TGLF's on average by 22%, 29%, and a factor of 2.9. Its relative spread (standard deviation over mean) is also smaller, 0.17, 0.14, and 0.26 against 0.25, 0.27, and 0.49. The surrogate is therefore biased upward and under-reacts to the input uncertainty at this operating point.

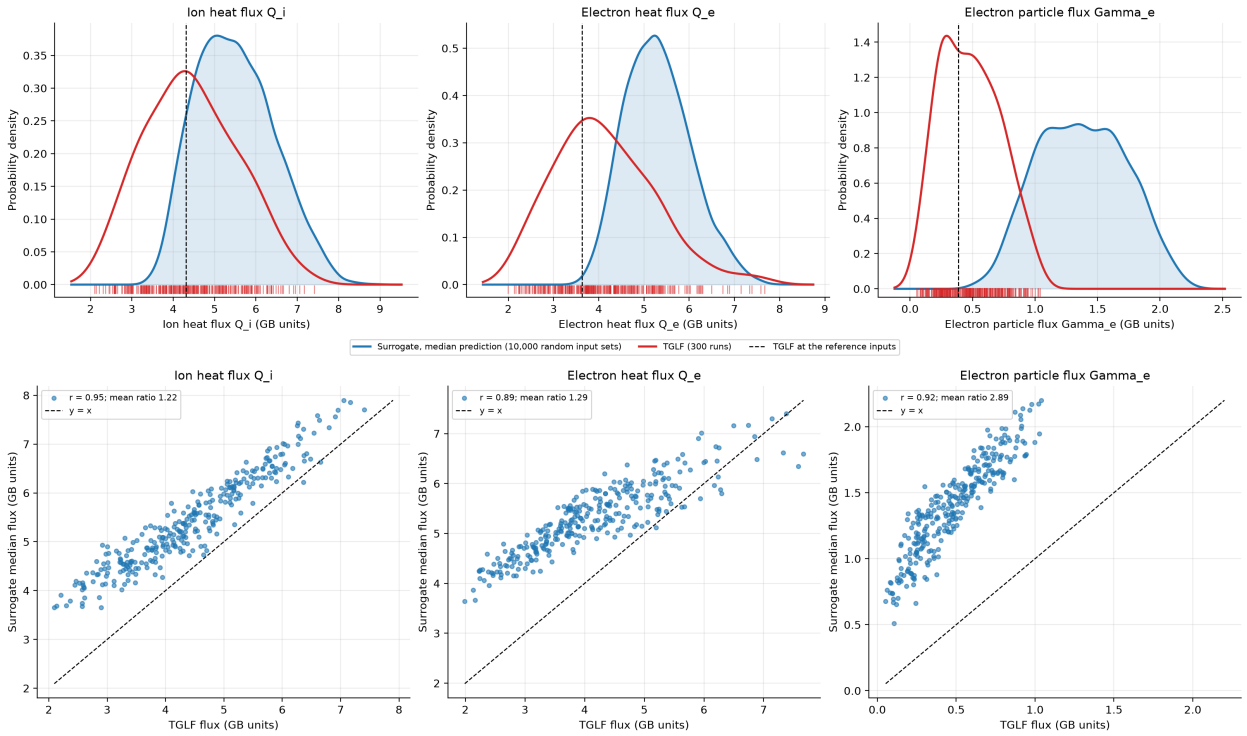


Figure 13. *Top: distribution of the flux under the input error bars for the surrogate (median prediction, 10,000 draws, blue) and TGLF (300 runs, red). The dashed line is TGLF at the reference inputs. Bottom: surrogate against TGLF on the same 300 input sets, with the line $y = x$. The legend gives the correlation and the mean ratio of surrogate to TGLF.*

The cause of this bias is not established, but the inputs give a hint. All 11 DIII-D inputs lie inside the training scan (Figure 14), but four are in its lowest quarter: the density gradient `RLNS_1` (8th percentile), the magnetic shear (11th), `RLTS_2` (22nd), and `Q_LOC` (23rd). The density gradient matters most for particle flux. It is the input most strongly linked to wider Γ_e intervals (Section 4.2), so the DIII-D case sits where the model is least certain about Γ_e.

To see why the surrogate reacts too little to the input errors, we changed one input at a time within its error bar and measured how much the flux moved (Table 9). The surrogate responds about half as strongly as TGLF to the density gradient. Its flux moves by 14.0% against 32.9% for Q_e and by 34.4% against 68.0% for Γ_e. With all inputs varied together, the surrogate's spread is one half to two thirds of TGLF's. TGLF was only run with all inputs varying together, so its per-input values are estimated from a straight-line fit to the 300

runs, which explains 88% to 91% of the variation (R^2 of 0.91, 0.88, and 0.91). They are approximate.

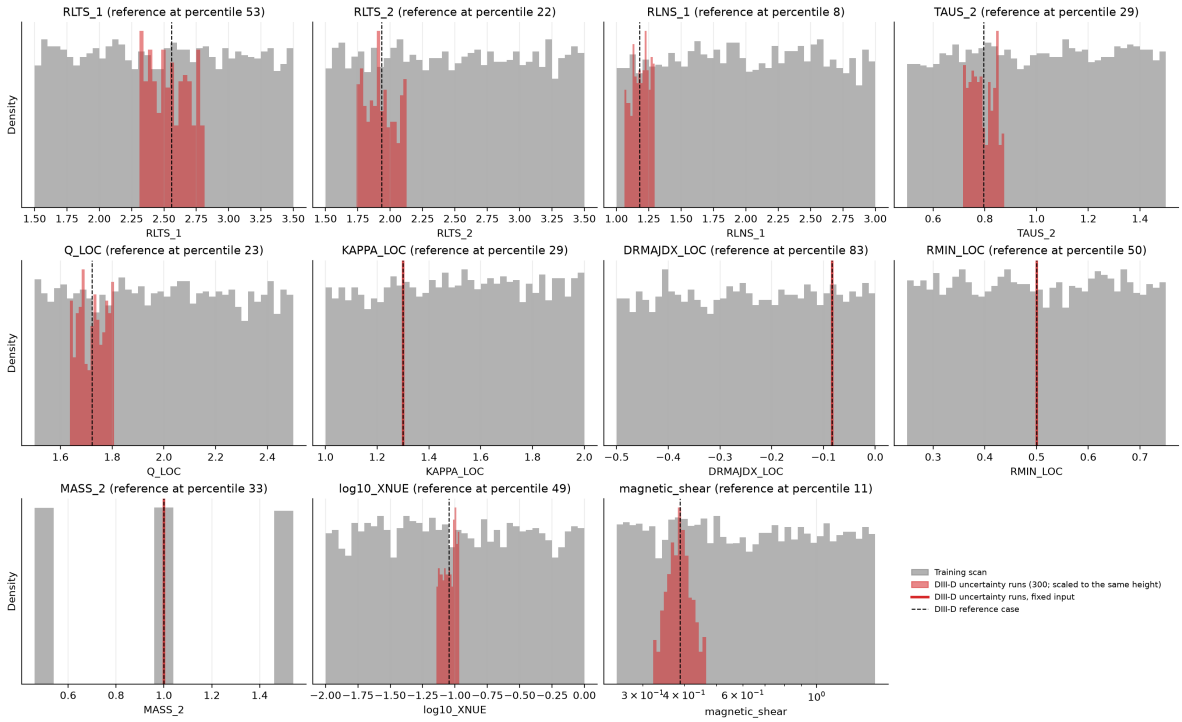


Figure 14. *Distribution of each model input in the training scan (grey) and in the 300 DIII-D uncertainty runs (red, scaled to the same height). The dashed line is the DIII-D reference case and the title gives its percentile within the training scan. Inputs fixed in the DIII-D runs appear as a single red line.*

	Surrogate			TGLF (linear fit)		
Varied input	Q_i	Q_e	Γ_e	Q_i	Q_e	Γ_e
`RLNS_1`	5.7	14.0	34.4	10.9	32.9	68.0
`RLTS_2`	21.5	6.7	11.3	31.5	3.7	8.2
All inputs together	28.0	22.9	41.4	42.3	44.8	80.0

Table 9. *Flux uncertainty (half of the 5–95% range, as a percentage of the mean flux) when one input has its error bar, and when all do. For TGLF, the per-input rows come from a straight-line fit to the 300 runs and the last row from the runs themselves.*

4.5 Member-Disagreement Intervals

The sections above report the quantile ensemble (Section 2.7). This subsection reports the 15-network ensemble (Section 2.8), whose intervals come from member disagreement.

On the test set its coverage was 91.2%, 90.1%, and 91.9% for Q_i, Q_e, and Γ_e. This is closer to the nominal 90% than the quantile intervals (Table 8), but it required multipliers fitted on validation data, which the quantile intervals did not.

Its intervals also failed on some inputs. In the outside-range collisionality group, ion-heat RMSE was 1.324 times the ordinary test value while the member standard deviation was only 0.343 times as large, so the members agreed more as the error grew. On the same 300 DIII-D input cases, its mean predicted particle flux was 0.995 against TGLF's 0.483, the same direction of bias as the quantile ensemble, despite 100% interval coverage. The average interval width was 3.85 in gyro-Bohm units. Broad intervals can contain the reference values while the central prediction stays biased. The cause of this local bias has not been established.

4.6 Gradient-Boosted Bootstrap Ensemble: Coverage, Extrapolation, and DIII-D Propagation

This subsection reports the 10-member bootstrap ensemble over the gradient-boosted histogram-tree surrogate (Section 2.9), a third, separately calibrated interval method evaluated on its own `scan_db` split (Section 3.6).

On its 750 in-distribution test cases, the calibrated 90% interval's empirical coverage was 91.20% for Q_i, 87.87% for Q_e, and 86.80% for Γ_e against a nominal target of 90%, close to nominal for Q_i but a few points low for Q_e and Γ_e. Predicted uncertainty correlated positively with absolute error (Pearson $r = 0.662$, 0.611, and 0.683 for Q_i, Q_e, and Γ_e), so wider bootstrap spread does track larger error on held-out in-distribution cases, with the same caveat as Section 4.2 that this is a loose, not precise, relationship.

We then evaluated the same ensemble, without retraining, on the nine `scan_db_outside` sweeps described in Section 3 and Table 1. Error grew in eight of the nine groups relative to the matched in-distribution test error, with the largest growth for Q_i under `ood_RLTS_2` (R/L_{Ti} swept to 3.5–5.0; mean absolute error rose from 16.80 to 83.94, a 5.00× increase) and for Γ_e under `ood_RLN` (R/L_n swept to 3.0–4.5; a 6.76× increase). The sole exception was `ood_XNUE`, where error fell below the in-distribution value for all three fluxes, consistent with high collisionality suppressing trapped-electron-mode and drift-wave turbulence toward zero transport. Ensemble spread grew in the same direction but by much less than the error: for example, under `ood_RLTS_2` the Q_i error ratio was 5.00 while the corresponding standard-deviation ratio was only 1.75. This mismatch has a structural cause. Because decision trees partition the input space using axis-aligned cuts at the training data's boundaries, a tree ensemble's predictions extrapolate as piecewise-constant functions once a query falls outside the training bounding box: every member keeps returning the value of whichever boundary leaf the query lands in, so bootstrap resampling changes which leaf is selected far less than it would change the true, unbounded physical flux. As a direct consequence, calibrated 90% coverage degraded substantially under extrapolation, falling to 53.3% for Q_i under `ood_RLTS_2` and to 66.7% for Q_i and Γ_e under `ood_DRMAJDX_LOC` (Δ' swept to -0.75–-0.5), the two lowest values among the 27 group-by-flux combinations. The error bars widen on these out-of-range inputs, giving a weak signal of elevated risk, but not enough to flag that the predictions have become unreliable.

Finally, we propagated the same DIII-D reference-case input error bars used in Section 4.4 through this ensemble with a 10,000-draw Monte Carlo sample, evaluated in 0.876 s total (87.6 μs per draw, about 11,400 draws per second on one CPU core). The resulting Q_i distribution had mean 3.56 and standard deviation 1.56 (a 43.7% relative spread, 90% range $[1.20, 6.33]$); Q_e had mean 5.21 and standard deviation 1.08 (20.8% relative spread, range $[3.46, 7.05]$); and Γ_e had mean 0.55 and standard deviation 0.20 (35.5% relative spread, range $[0.24, 0.88]$). Measured against the 300 ground-truth TGLF runs at the same input error bars, the surrogate's point predictions correlated well with TGLF ($r = 0.891$, 0.687, and 0.742 for Q_i, Q_e, and Γ_e), and the calibrated 90% interval around the matched 300-case predictions contained the TGLF value in 100.0%, 99.7%, and 99.7% of cases, wider than the nominal 90% target. A single-input sensitivity sweep, varying one input at a time within its own error bar while holding the rest at the exact reference values (Table 10), shows that the dominant driver differs by flux: R/L_{Ti} (`RLTS_2`) dominates the Q_i spread, collisionality ν_{ei} (`XNUE`) dominates the Q_e spread, and the density gradient R/L_n (`RLNS`) dominates the Γ_e spread.

Varied input (error bar)	Std(Q_i)	Std(Q_e)	Std(Γ_e)
`RLTS_1` (±10%)	0.228	0.532	0.083
`RLTS_2` (±10%)	1.191	0.365	0.071
`RLNS` (±10%)	0.146	0.443	0.137
`TAUS_2` (±10%)	0.401	0.154	0.019
`XNUE` (±20%)	0.752	0.651	0.024
`Q_LOC` (±5%)	0.289	0.325	0.093
`Q_PRIME_LOC` (±10%)	0.132	0.095	0.025

Table 10. *Standard deviation induced in each flux (gyro-Bohm units) when a single DIII-D input is varied within its own measurement error bar (5,000 draws), holding all other inputs at their exact reference values. The largest entry in each column is the dominant single-input driver for that flux.*

5 Discussion, Limitations, and Remaining Work

These experiments show that representation and training choices can matter as much as model complexity. Logarithmic collision frequency improved the small-data linear fits. On the larger database, the MLP had the lowest error and the fastest prediction of the models we compared, and the accuracy ensemble improved on its baselines, although activation changes traded accuracy across outputs. The uncertainty results add three findings. Interval width follows how much training data is available, so the intervals show where the surrogate is reliable. Error rises on inputs outside the training range, but a narrow interval does not guarantee accuracy. And at the DIII-D reference case the surrogate predicts higher fluxes than TGLF and responds less strongly to the input errors. Sign errors, incorrect trends, and this local bias show why average error and interval coverage alone are insufficient.

In practice, these results support using the surrogate for inputs inside the dense part of the `scan_db` range, where its intervals are narrow and its error tends to be lower. Its predictions need checking against TGLF for low-flux conditions, for inputs outside the training range, and for conditions such as the DIII-D reference case that were not part of the training scan. A wide interval is a useful warning in these cases, but a narrow one is not proof that the prediction is accurate.

The conclusions remain limited by the small structured starter set, exploratory model-family selection, prior test inspection, and reuse of validation data for uncertainty calibration. The learning curves cover one nested sequence. Because the experiments use different datasets and procedures, they cannot establish one model ranking or determine whether ensembles outperform the group's boosted trees or a matched single network.

5.1 Planned Model-Comparison Experiments

We intend to test whether expanding the ensemble improves prediction enough to justify the additional training and inference cost. The 32-network model described by Meneghini et al. (2017) motivates studying member count as a design choice. A controlled comparison will hold the architecture, training data, and fitting procedure fixed while varying the number of members, including a single-network reference. Our existing accuracy, quantile, and 15-network ensembles (Table 2) do not isolate this effect, because they differ in more than member count.

Other proposed experiments concern alternative transformations of the input features, different network depths and widths, and separate ensembles for Q_i, Q_e, and Γ_e in place of the current shared three-output networks. The input-transform comparisons will extend beyond the current logarithmic collision-frequency representation. Learned transformations will be fitted on training data only. These choices will be selected using validation data and evaluated for each target, with computational cost reported for the full predictor. We will consolidate the group's other models under the same evaluation procedure and preserve new assessment cases from further tuning.

5.2 Planned DIII-D Database Study

Our mentor's November 2 project plan provides a sequence for applying these methods to a DIII-D database. The first stage is to run TGLF at 13 supplied radial locations and verify the results against the reference answers. The Task 4 handout then specifies 500–1,000 additional cases per team member, spread across these radii. Five inputs are varied around each reference condition: the two temperature gradients, shared density gradient, temperature ratio, and collision frequency. The database will retain the prescribed geometry at each radius and apply the stated cleaning rule for negative heat-flux cases.

The next stages are to train a baseline and neural network for the three fluxes, attach uncertainty estimates, and evaluate new input cases supplied by the mentors. Learning curves using 500, 1,000, 2,000, and all available training cases will examine whether errors con-

tinue to decrease, intervals shrink, and some input regions remain poorly represented. The final comparison will evaluate the broad `scan_db` surrogate and the DIII-D-trained surrogate on the same DIII-D test cases, assessing both accuracy and the reliability of uncertainty estimates.

The plan also lists two possible extensions: comparing surrogate error with the discrepancy between TGLF and nonlinear CGYRO outputs, and examining the contributions of low- and high-wavenumber turbulence to electron heat flux. These extensions are separate from the main comparison. Throughout the study, the plan emphasizes correlated plasma inputs, data quality, uncertainty estimates for every prediction, and flags for inputs outside the training data. This work remains distinct from the completed experiments reported above.

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